NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

Affiliated to Pokhara University

LAB REPORT FOR SIMULATION AND MODELING

notes.arpanadhikari7.com.np

LAB REPORT 1: VERIFICATION OF OHM'S LAW

Submitted by:
Name: Arpan Adhikari
Roll No.: 231309
BECE (6th SEM)

Submitted to:
Resha Deo

Available on notes.arpanadhikari7.com.np

THEORY:

Ohm's Law is one of the fundamental principles of electrical engineering and physics. It was proposed by Georg Simon Ohm in 1827. The law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided that the temperature and other physical conditions remain constant.
The mathematical relationship is:
V = I × R
Rearranging the equation:
I = V / R
R = V / I
Where:

- **V** = Voltage (Volts, V)
- **I** = Current (Amperes, A)
- **R** = Resistance (Ohms, Ω)

Graphical Interpretation

According to Ohm's Law, for a conductor with constant resistance, voltage and current are directly proportional. When Voltage (V) is plotted on the X-axis and Current (I) on the Y-axis, the resulting graph is a straight line passing through the origin.
The slope of the graph is:
Slope = ΔI / ΔV = 1 / R
The graph provides useful information about the resistance of the conductor:

- Higher resistance → Smaller slope (less steep line)
- Lower resistance → Larger slope (steeper line)
- A straight-line graph passing through the origin indicates that the conductor obeys Ohm's Law

Simulation Using C File Handling

In this experiment, a software-based simulation is used instead of a physical circuit. A C program is developed to calculate current values for different voltages using Ohm's Law and store the results in a CSV file.
The simulation performs the following tasks:

notes.arpanadhikari7.com.np

- Varies the voltage from 0 V to 20 V in steps of 1 V.
- Calculates the corresponding current using:

I = V / R

- Stores the voltage and current values in a CSV file.
- Uses C file handling functions to create and manage the output file.

Function	Purpose
fopen()	Opens a file for reading or writing
fprintf()	Writes formatted data to a file
fclose()	Closes the file and saves all data

Available on notes.arpanadhikari7.com.np

C-PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>

#define RESISTANCE    100.0
#define VOLTAGE_MIN     0.0
#define VOLTAGE_MAX    20.0
#define VOLTAGE_STEP    1.0
#define OUTPUT_FILE  "ohms_law.csv"

int main()
{
   FILE   *fp;
   double  voltage, current;
   fp = fopen(OUTPUT_FILE, "w");
   if (fp == NULL)
   {
      printf("Error opening file!\n");
      return 1;
   }
   fprintf(fp, "Voltage (V),Current (A),Resistance (Ohm)\n");
   for (voltage = VOLTAGE_MIN;
        voltage <= VOLTAGE_MAX;
        voltage += VOLTAGE_STEP)
   {
      current = voltage / RESISTANCE;
      fprintf(fp, "%.2f,%.4f,%.2f\n",
            voltage, current, RESISTANCE);
   }
   fclose(fp);
   printf("Data written to %s\n", OUTPUT_FILE);
   return 0;
}
```

OBSERVATIONS AND DATA TABLE:

The following table shows the simulated data generated by the C program for a fixed resistance of 100 Ω:

notes.arpanadhikari7.com.np

Voltage (V)	Current (A)	Resistance (Ω)
0.00	0.0000	100.00
1.00	0.0100	100.00
2.00	0.0200	100.00
3.00	0.0300	100.00
4.00	0.0400	100.00
5.00	0.0500	100.00
10.00	0.1000	100.00
15.00	0.1500	100.00
20.00	0.2000	100.00

Available on notes.arpanadhikari7.com.np

The graph below shows the V-I characteristic curve generated from the simulated data. The straight line through the origin confirms Ohm's Law.

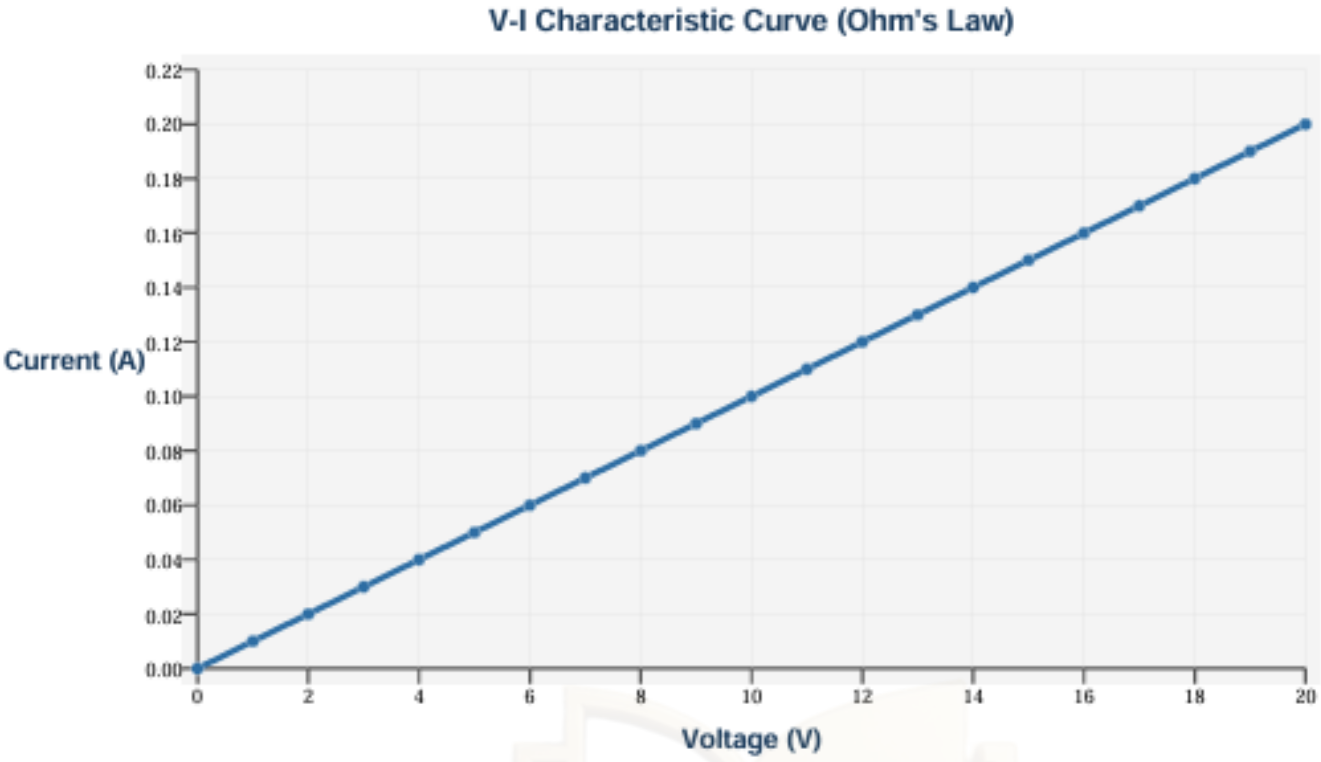


Figure 1: V-I Characteristic Curve. The slope of the line = 1/R = 0.01 A/V.

RESULT:

The simulation verified Ohm's Law successfully. The current increased linearly with voltage, confirming a direct proportional relationship. The V–I graph obtained was a straight line passing through the origin. The calculated slope matched the theoretical value:
Slope = 1 / R = 1 / 100 = 0.01 A/V
The CSV file successfully stored all generated voltage and current values for analysis.

CONCLUSION:

notes.arpanadhikari7.com.np

This experiment demonstrated the verification of Ohm's Law using a C programming simulation and file handling. The results showed that current is directly proportional to voltage and the V–I graph forms a straight line through the origin. The theoretical slope value was validated, confirming the accuracy of the simulation.
Hence, Ohm's Law was successfully verified using C programming.

Available on notes.arpanadhikari7.com.np